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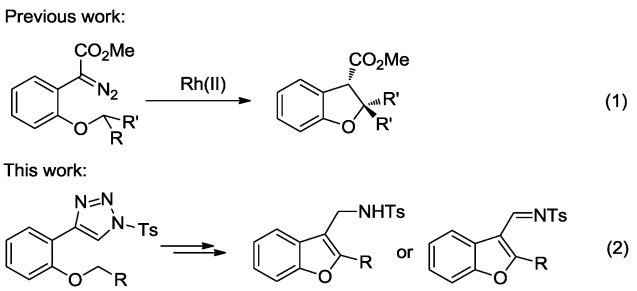
One-pot synthesis of 2,3-disubstituted dihydrobenzofurans and benzofurans *via* rhodium-catalyzed intramolecular C–H insertion reaction†

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Intramolecular sp^3 C–H insertion reaction of α-imino rhodium carbene generated from *N*-sulfonyl-1,2,3-triazoles has been described. A number of 2,3-dihydrobenzofuran and benzofuran derivatives have been obtained in good to excellent yields.

Benzofuran and dihydrobenzofuran are versatile building blocks in organic synthesis and key structures of numerous natural products with biological activities.[1] Numerous efforts have been made to construct benzofuran and dihydrobenzofuran skeletons. Several classic strategies are available including intramolecular Friedel-Crafts reaction and Wittig reaction.[2] Moreover, transition-metal-catalyzed C–C and C–O formation reactions represent atom-economic protocols for the construction of benzofuran moieties.[3] For instance, 2,3-disubstituted benzofurans can be synthesized from *o*-functionalized phenols and their derivatives through palladium-, copper-, platinium- and silver-catalyzed annulation reactions.[3a–g] Metal carbenoids, especially rhodium carbenoids, generated from diazo compounds, have been applied to the synthesis of dihydrobenzofuran motifs (Scheme 1, eqn (1)), but only limited examples were reported.[4] The need to develop novel and efficient approaches to synthesize benzofurans is still required.

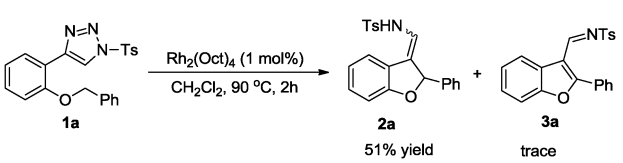
Recently, *N*-sulfonyl-1,2,3-triazoles have emerged as alternative precursors for the formation of metallocarbenes.[5] The α-imino carbenes derived from *N*-sulfonyl triazoles are capable of undergoing a variety of useful reactions, including the synthesis of N-heterocycles,[6] stereoselective cycloadditions,[7] C–H bond insertion,[6o,p,8] X–H bond insertion (X = heteroatoms),[9] and other transformations.[10] The C–H bond insertion reactions of α-imino carbenes leading to heterocyclic compounds and enamides were restricted to electron rich sp^2 C–H bonds.[6o,p,8b,c] So far only one example of sp^3 C–H insertion of α-imino rhodium carbenes was reported to give β-chiral sulfonamides.[8a] Herein, we wish to report the synthesis of 2,3-dihydrobenzofuran and benzofuran derivatives *via* the intramolecular sp^3 C–H insertion reactions of α-imino rhodium carbenes.



Scheme 1 Metal carbenoid strategy for synthesis of benzofurans.

Our initial studies began with the intramolecular C–H bond insertion reaction of 4-(2-(benzyloxy)phenyl)-1-tosyl-1*H*-1,2,3-triazole **1a**. Using $Rh_2(Oct)_4$ as the catalyst, ring-opening of the triazole and subsequent insertion of the α-imino carbene towards the C–H bond adjacent to oxygen took place leading to **2a** in 51% yield as a mixture of two *E*/*Z* isomers together with trace amounts of **3a** at 90 °C (Scheme 2).

Compound **2a** bearing an interesting exocyclic double bond and versatile functional groups might be utilized for rapid construction of molecular complexity. Approaches to synthesize such a skeleton were quite limited so far.[11] We optimized the reaction conditions. $Rh_2(OAc)_4$ and $Rh_2(OPiv)_4$ are also active



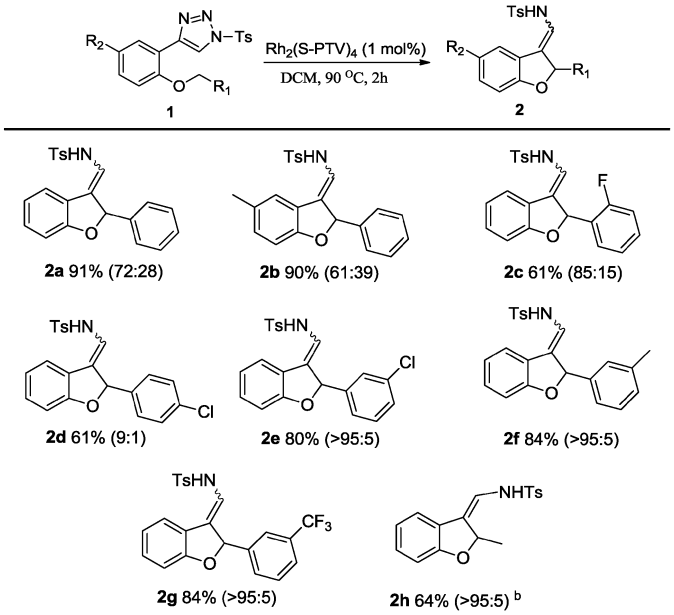
Scheme 2 Rh(II)-catalyzed intramolecular C–H bond insertion of **1a**.

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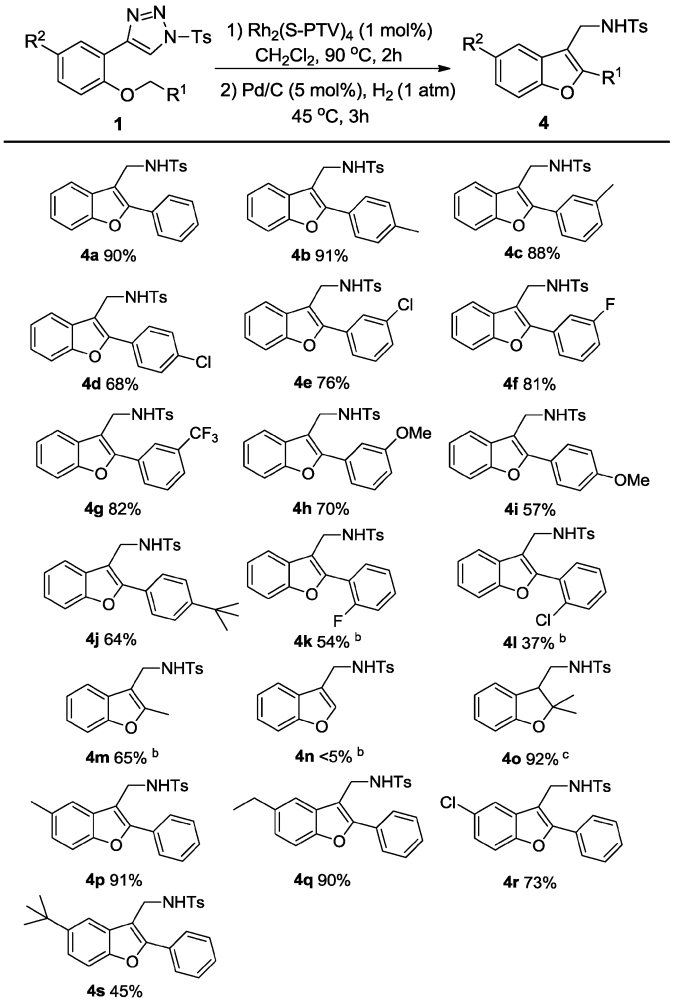
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Table 1 Synthesis of 2,3-dihydrobenzofuran derivatives **2**[a]



[a] Reaction conditions: **1** (0.5 mmol), Rh_2(S-PTV)$_4$ (1 mol%), CH_2Cl_2 (4 mL), 90 °C, 2 h, under N_2. [b] 4 h.

Table 2 Scope of the reaction for the synthesis of **4**[a]



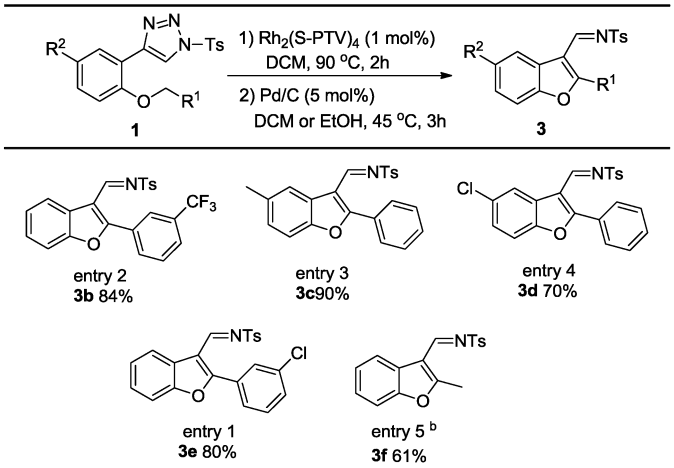
[a] Reaction conditions: under N_2, **1** (0.5 mmol), Rh_2(S-PTV)$_4$ (1 mol%) and CH_2Cl_2 (4 mL) were stirred in a sealed tube at 90 °C for 2 h. Then 5 mol% of Pd/C was added, and the mixture was stirred at 45 °C under H_2 for 3 h. [b] 4 h. [c] 70 °C.

giving **2a** in 30% and 57% yields. Electron-deficient Rh_2(TFA)$_4$ is totally inactive. Bulky ligands containing Hashimoto-type catalysts could significantly improve the reaction, and **2a** could be obtained in 91% yield in CH_2Cl_2 with 1 mol% of Rh_2(S-PTV)$_4$ as the catalyst (see Table S1 in the ESI†). CH_2Cl_2 is the most appropriate solvent. As demonstrated in Table 1, under the optimized conditions, **2a–2b** were isolated in *ca.* 90% yields as mixtures of two isomers with *E*/*Z* ratios of 72 : 28 and 61 : 39, respectively. Due to the steric effect, the *E*/*Z* ratio was increased to 85 : 15 for **2c**. In the cases of **1e–1g** bearing more steric substituents, almost pure (*E*)-isomers of **2e–2g** could be isolated after simple chromatography. Unexpectedly, only (*Z*)-isomer of **2h** was obtained in 64% yield.

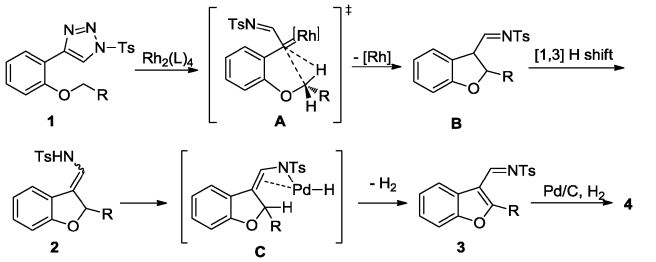
Interestingly, we found that catalytic hydrogenation of **2a** in the presence of 5 mol% of Pd/C catalyst and 1 atm H_2 at 45 °C did not give the expected 2,3-dihydrobenzofuran, instead, benzofuran **4a** was obtained almost quantitatively. Moreover, we found that **4a** could be obtained in a one-pot manner without isolation of **2a**. After completion of rhodium-catalyzed C–H insertion reaction of **1**, Pd/C was added to the mixture and switched to a H_2 atmosphere, **4a–s** were directly afforded in good to excellent yields. The results were summarized in Table 2. For substrates **1b–1c** containing a methyl at 3- or 4-postions of the benzyl groups, the reactions proceeded smoothly to give the corresponding products **4b** and **4c** in excellent yields (88 and 91%). However, when electron-withdrawing Cl was introduced at the *para*-position of the benzyl group, the yield of **4d** was sharply decreased to 68% due to the impaired electronic density. Similar results were observed when Cl, F, CF_3 and OMe were at *meta* positions of the benzyl groups. In the case of **1i** bearing a methoxy at the *para* position of the benzyl group, **4i** was obtained in only 57% yield, and obvious decomposition of the starting material was observed. However, the methoxy at the *meta* position has less influence.

Probably due to the steric effect, the *tert*-butyl group failed to increase the yield of **4j**. The steric effect was quite obvious when the benzyl groups containing F and Cl groups at their *ortho* positions, and the corresponding products **4k** and **4l** were afforded in 54 and 37% yields, respectively, even the reaction time was elongated to 4 hours. The intramolecular C–H insertion of **1m** and **1n** were also studied. The corresponding benzofuran **4m** was delivered in moderate yield, but the reaction of **1n** was sluggish to give the desired product in a very poor yield. The results indicate that the primary C–H bond is much inert than the secondary C–H bond and benzyl C–H bonds. In sharp contrast, **4o** was obtained in excellent yield even the reaction temperature was lowered to 70 °C. Under the same conditions, other substrates containing additional methyl, ethyl and Cl groups on the phenyl ring gave corresponding benzofurans **4p–4r** in good to excellent yields.

Table 3 Synthesis of benzofuranyl imines[a]



[a] Reaction conditions: under N_2, **1** (0.5 mmol), Rh_2(S-PTV)$_4$ (1 mol%) and CH_2Cl_2 (4 mL) were stirred in a sealed tube at 90 °C for 2 h. Then 5 mol% of the Pd/C catalyst was added, and the mixture was stirred at 45 °C for 3 h. [b] 4 h.

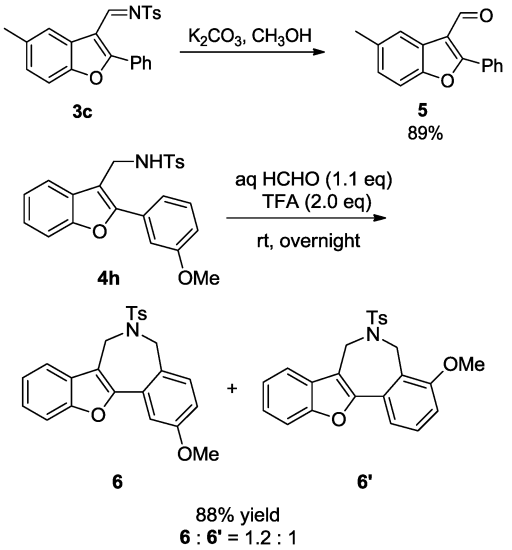


Scheme 3 Proposed catalytic pathway of **4**.

However, a bulky *tert*-butyl group retarded the reaction and only 45% yield was obtained.

The formation of **4** appears to proceed *via* the insertion of α-imino rhodium carbene towards a C–H bond adjacent to oxygen and subsequent isomerization of the double bond in the presence of a Pd/C catalyst. However, when the reaction was carried out under a N_2 atmosphere instead of H_2, the isolated **2** was transformed into benzofuranyl imine **3** rather than **4** in the presence of Pd/C in CH_2Cl_2 or EtOH. This indicates that the transformation of **2** to **4** is not a simple isomerization process. Similarly, **3** could also be obtained from **1** *via* the one pot, two-step process in a N_2 atmosphere. The results are summarized in Table 3. The overall yields depend on the carbene insertion reaction of C–H bonds.

A proposed catalytic pathway for the formation of **4** is outlined in Scheme 3. First, reaction of triazole **1** with the Rh(II) complex would generate α-imino rhodium carbene species, and intramolecular C–H insertion would give 2,3-dihydrofuran **B** *via* a three-centered transition state **A**.[12] The imine **B** is not stable, and a rapid [1,3]-H shift would occur giving more stable **2**. We assumed that **4** was formed *via* palladium-catalyzed dehydrogenation and hydrogenation. Oxidative addition of sulfonamide N–H would produce Pd–H species **C**.[13] After reductive elimination, **3** is formed. Finally, hydrogenation of **3** could give **4**.



Scheme 4 Further transformation of **3c** and **4h** into aldehyde and azepine.

The reactive imino groups of compound **3** can be transformed into other functional groups (Scheme 4). Treatment of **3c** with K_2CO_3 in methanol at room temperature afforded benzofuran-3-carbaldehyde **5**. Moreover, treatment of **4h** with formaldehyde in dichloromethane at room temperature could give tetracyclic azepine **6** in 88% yield. Regioisomers of **6** and **6′** were isolated in a ratio of 1.2 : 1. The present strategy offers a rapid synthetic route of polycyclic azepines in excellent yields. Further effort is required to increase the regioselectivity of the products.

In conclusion, we have described a facile route for the synthesis of 2,3-dihydrobenzofurans and benzofurans. The reactions proceed *via* rhodium-catalyzed ring-opening of *N*-sulfonyl-1,2,3-triazles, the intramolecular sp^3 C–H insertion reaction of α-imino rhodium carbene, and subsequent palladium-catalyzed dehydrogenation. 2,3-Disubstituted hydrobenzofurans and benzofurans can be obtained in good to excellent yields in a one-pot manner.

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Notes and references

1 (*a*) Y. H. Liu, M. Kubo and Y. Fukuyama, *J. Nat. Prod.*, 2012, **75**, 2152; (*b*) M. H. Sun, C. Zhao, G. A. Gfesser, C. Thiffault, T. R. Miller and M. Cowart, *J. Med. Chem.*, 2005, **48**, 6482; (*c*) T. Vang, Y. L. Xie, Y. D. Liu, S. D. Wu and L. Tautz, *J. Med. Chem.*, 2011, **54**, 562; (*d*) M. D. Argade, A. Y. Mehta, A. Sarkar and U. R. Desai, *J. Med. Chem.*, 2014, **57**, 3559; (*e*) N. Negoro, S. Sasaki, S. Mikami, M. Ito and T. Yasuma, *J. Med. Chem.*, 2012, **55**, 3960; (*f*) T. Nagahara, T. Yokoyama, K. Inamura and M. Iwamoto, *J. Med. Chem.*, 1994, **37**, 1200; (*g*) Z. Yang, H. B. Liu, C. M. Lee, H. M. Chang and H. N. C. Wong, *J. Org. Chem.*, 1992, **57**, 7248.

2 (*a*) J. M. Lee, M. Kim and I. Kim, *J. Org. Chem.*, 2014, **79**, 6153; (*b*) S. Syu, Y. T. Lee, Y. J. Jang and W. W. Lin, *Org. Lett.*, 2011, **13**, 2970; (*c*) C. R. Reddy, G. Krishna, N. Kavitha, B. Latha and D. S. Shin, *Eur. J. Org. Chem.*, 2012, 5381.

3 (*a*) H. Yuan, K. J. Bi, B. Li, H. Z. Jin, Q. Y. Sun and W. D. Zhang, *Org. Lett.*, 2013, **15**, 4742; (*b*) S. Q. Ye, G. Liu, S. Z. Pu and J. Wu, *Org. Lett.*, 2012, **14**, 70; (*c*) X. W. Guo, R. Yu, H. J. Li and Z. P. Li, *J. Am. Chem. Soc.*, 2019, **131**, 17387; (*d*) I. Nakamura, Y. Mizushima and Y. Yamamoto, *J. Am. Chem. Soc.*, 2005, **127**, 15022; (*e*) B. Gabriele, R. Mancuso, G. Salerno and L. Veltri, *Chem. Commun.*, 2005, 271; (*f*) B. Gabriele, R. Mancuso, G. Salerno and M. Costa, *Adv. Synth. Catal.*,

 This journal is © The Royal Society of Chemistry 2015

2006, **348**, 1101; (*g*) B. Gabriele, R. Mancuso, E. Lupinacci, G. Salerno and L. Veltri, *Tetrahedron*, 2010, **66**, 6156; (*h*) B. Anxionnat, D. G. Pardo, G. Ricci, K. Rossen and J. Cossy, *Org. Lett.*, 2013, **15**, 3876; (*i*) N. Wongsa, U. Sommart, T. Ritthiwigrom, A. Yazici, S. Kanokmedhakul, A. C. Willis and S. G. Pyne, *J. Org. Chem.*, 2013, **78**, 1138; (*j*) C. R. Liu, M. B. Li, C. F. Yang and S. K. Tian, *Chem. – Eur. J.*, 2009, **15**, 793.

4 (*a*) H. M. L. Davies, M. V. A. Grazini and E. Aouad, *Org. Lett.*, 2001, **3**, 1475; (*b*) H. Saito, H. Oishi, S. Kitagaki, S. Nakamura and S. Hashimoto, *Org. Lett.*, 2002, **4**, 3887; (*c*) R. P. Reddy, G. H. Lee and H. M. L. Davies, *Org. Lett.*, 2006, **8**, 3437; (*d*) C. Soldi, K. N. Lamb and J. T. Shaw, *J. Am. Chem. Soc.*, 2014, **136**, 15142.

5 (*a*) T. Horneff, S. Chuprakov, N. Chernyak, V. Gevorgyan and V. V. Fokin, *J. Am. Chem. Soc.*, 2008, **130**, 14972; (*b*) S. Chuprakov, F. W. Hwang and V. Gevorgyan, *Angew. Chem., Int. Ed.*, 2007, **46**, 4757; (*c*) H. M. L. Davies and J. S. Alford, *Chem. Soc. Rev.*, 2014, **41**, 5151; (*d*) B. Chattopadhyay and V. Gevorgyan, *Angew. Chem., Int. Ed.*, 2012, **51**, 862.

6 (*a*) F. Medina, C. Besnard and J. Lacour, *Org. Lett.*, 2014, **16**, 3232; (*b*) T. Miura, M. Yamauchi and M. Murakami, *Chem. Commun.*, 2009, 1470; (*c*) B. Chattopadhyay and V. Gevorgyan, *Org. Lett.*, 2011, **13**, 3746; (*d*) B. T. Parr, S. A. Green and H. M. L. Davies, *J. Am. Chem. Soc.*, 2013, **135**, 4716; (*e*) J. E. Spangler and H. M. L. Davies, *J. Am. Chem. Soc.*, 2013, **135**, 6802; (*f*) M. Zibinsky and V. V. Fokin, *Angew. Chem., Int. Ed.*, 2013, **52**, 1507; (*g*) S. Chuprakov, S. W. Kwok and V. V. Fokin, *J. Am. Chem. Soc.*, 2013, **135**, 4652; (*h*) E. E. Schultz and R. Sarpong, *J. Am. Chem. Soc.*, 2013, **135**, 4696; (*i*) T. Muira, T. Biyajima, T. Fujii and M. Murakami, *J. Am. Chem. Soc.*, 2012, **134**, 194; (*j*) T. Miura, K. Hiraga, T. Biyajima, T. Nakamuro and M. Murakami, *Org. Lett.*, 2013, **15**, 3298; (*k*) J. S. Alford, J. E. Spangler and H. M. L. Davies, *J. Am. Chem. Soc.*, 2013, **135**, 11712; (*l*) Y. Shi and V. Gevorgyan, *Org. Lett.*, 2013, **15**, 5394; (*m*) X. J. Ma, S. F. Pan, H. X. Wang and W. Z. Chen, *Org. Lett.*, 2014, **16**, 4554; (*n*) C. E. Kim, S. Park, D. Eom, B. Seo and P. H. Lee, *Org. Lett.*, 2014, **16**, 1900; (*o*) J. M. Yang, C. Z. Zhu, X. Y. Tang and M. Shi, *Angew. Chem., Int. Ed.*, 2014, **53**, 5142; (*p*) B. Rajagopal, C. H Chou, C. C. Chung and P. C. Lin, *Org. Lett.*, 2014, **16**, 3752; (*q*) X. Y. Tang, Y. S. Zhang, L. He, Y. Wei and M. Shi, *Chem. Commun.*, 2015, **51**, 133; (*r*) D. J. Lee, H. S. Han, J. Shin and E. J. Yoo, *J. Am. Chem. Soc.*, 2014, **136**, 11606.

7 (*a*) S. Chuprakov, S. W. Kwok, L. Zhang, L. Lercher and V. V. Fokin, *J. Am. Chem. Soc.*, 2009, **131**, 18034; (*b*) N. Grimster, L. Zhang and V. V. Fokin, *J. Am. Chem. Soc.*, 2010, **132**, 2510; (*c*) S. W. Kwok, L. Zhang, N. P. Grimster and V. V. Fokin, *Angew. Chem., Int. Ed.*, 2014, **53**, 3452; (*d*) T. Miura, Y. Funakoshi and M. Murakami, *J. Am. Chem. Soc.*, 2014, **136**, 2272; (*e*) B. T. Parr and H. M. L. Davies, *Angew. Chem., Int. Ed.*, 2013, **52**, 10044; (*f*) J. S. Alford and H. M. L. Davies, *Org. Lett.*, 2012, **14**, 6020.

8 (*a*) S. Chuprakov, J. A. Malik, M. Zibinsky and V. V. Fokin, *J. Am. Chem. Soc.*, 2011, **133**, 10352; (*b*) D. Yadagiri and P. Anbarasan, *Org. Lett.*, 2014, **16**, 2510; (*c*) S. Park, W. S. Yong, S. Kim and P. H. Lee, *Org. Lett.*, 2014, **16**, 4468.

9 (*a*) T. Miura, T. Biyajima, T. Fujii and M. Murakami, *J. Am. Chem. Soc.*, 2012, **134**, 194; (*b*) T. Miura, T. Tanaka, T. Biyajima, A. Yada and M. Murakami, *Angew. Chem., Int. Ed.*, 2013, **52**, 3883; (*c*) S. Chuprakov, B. T. Worrell, N. Selander, R. K. Sit and V. V. Fokin, *J. Am. Chem. Soc.*, 2014, **136**, 195; (*d*) T. Miura, T. Tanaka, A. Yada and M. Murakami, *Chem. Lett.*, 2013, 1308.

10 (*a*) D. Yadagiri and P. Anbarasan, *Chem. – Eur. J.*, 2013, **19**, 15115; (*b*) R. Liu, M. Zhang, G. W. M. Pherson and W. Tang, *Chem. Commun.*, 2013, **49**, 4376; (*c*) T. Miura, Y. Funakoshi, M. Morimoto, T. Biyajima and M. Murakami, *J. Am. Chem. Soc.*, 2012, **134**, 17440; (*d*) N. Selander, B. T. Worrell and V. V. Fokin, *Angew. Chem.*, 2012, **124**, 13231; (*e*) A. Boyer, *Org. Lett.*, 2014, **16**, 1660; (*f*) D. J. Jung, H. J. Jeon, J. H. Kim, Y. Kim and S. Lee, *Org. Lett.*, 2014, **16**, 2208; (*g*) T. Miura, T. Nakamuro and M. Murakami, *J. Am. Chem. Soc.*, 2014, **136**, 15905; (*h*) N. Selander, B. T. Worrell, S. Chuprakov, S. Velaparthi and V. V. Fokin, *J. Am. Chem. Soc.*, 2014, **134**, 14670.

11 (*a*) Y. Terada, M. Arisawa and A. Nishida, *Angew. Chem., Int. Ed.*, 2004, **43**, 4063; (*b*) M. Arisawa, Y. Terada, K. Takahashi, M. Nakagawa and A. Nishida, *J. Org. Chem.*, 2006, **71**, 4255; (*c*) K. Sasaki, Y. Kondo and K. Maruoka, *Angew. Chem., Int. Ed.*, 2001, **40**, 411; (*d*) A. L. Girard, R. Lhermet, C. Fressigne, M. Durandetti and J. Maddaluno, *Eur. J. Org. Chem.*, 2012, 2895; (*e*) M. Durandetti, L. Hardou, R. Lhermet, M. Rouen and J. Maddaluno, *Chem. – Eur. J.*, 2011, **17**, 12773.

12 (*a*) E. Nakamura, N. Yoshikai and M. Yamanaka, *J. Am. Chem. Soc.*, 2002, **124**, 7181; (*b*) H. M. L. Davies and A. R. Dick, *Top. Curr. Chem.*, 2010, **292**, 303.

13 (*a*) J. Broggi, O. Songis, A. Poater, L. Cavallo, A. M. Z. Slawin and C. S. J. Cazin, *J. Am. Chem. Soc.*, 2013, **135**, 4588; (*b*) T. Ohmura, A. Kijima and M. Suginome, *J. Am. Chem. Soc.*, 2009, **131**, 6070.